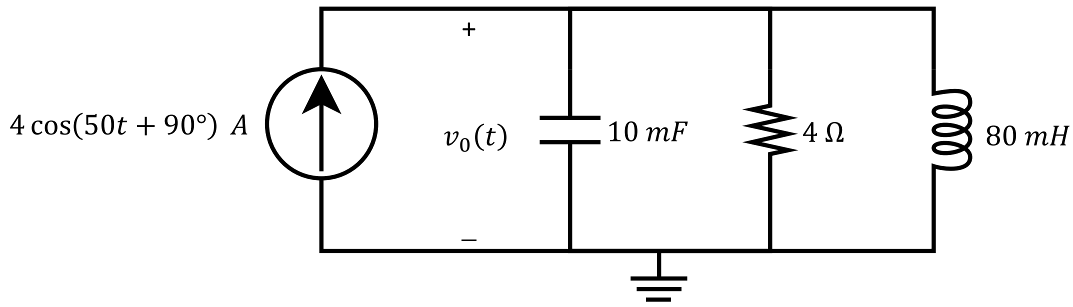
EBN111 CLASS TEST 3 (26 MAY 2014)

Vraag 1 [2]	Question 1 [2]
Bepaal $v_0(t)$. (Skryf antwoord as sinusied met positiewe ampitude en fasehoek tussen ±180°.)	Determine $v_0(t)$. (Write answer as a sinusoid with positive amplitude and phase angle between ±180°.)



<table>
<tr>
<td>Vraag 2 [3]
Bepaal die ekwivalente impedansie in terme van terminale $a - b$ vir beide kringbane. Skryf asseblief die antwoorde in reghoekige formaat.</td>
<td>Question 2 [3]
Determine the equivalent impedance with respect to terminals $a - b$ for both circuits. Please write answers in rectangular format.</td>
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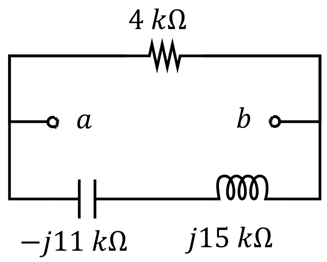
<td>Kringbaan / Circuit (a) [1]



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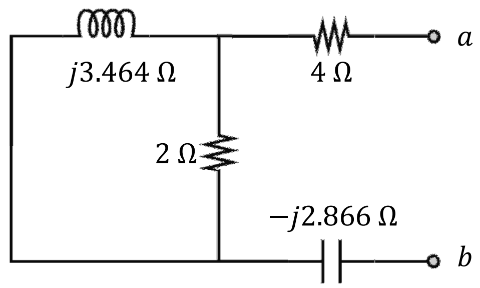
<td>Kringbaan / Circuit (b) [2]



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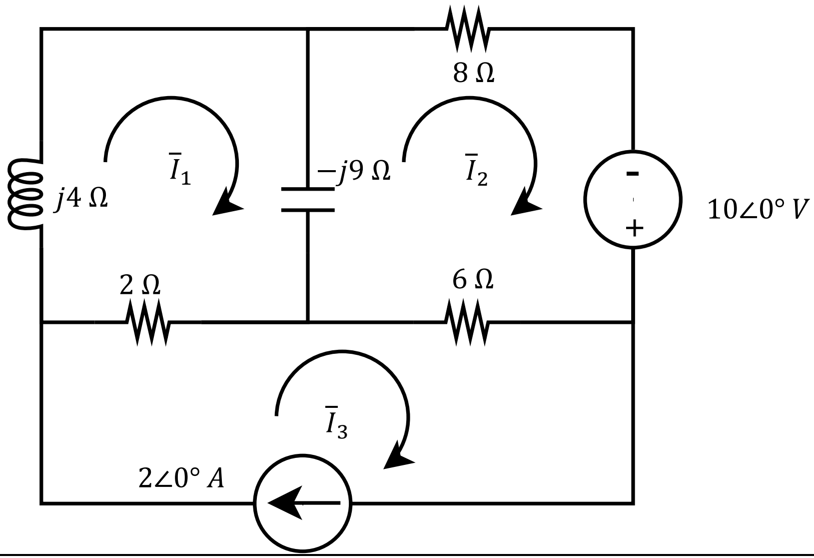
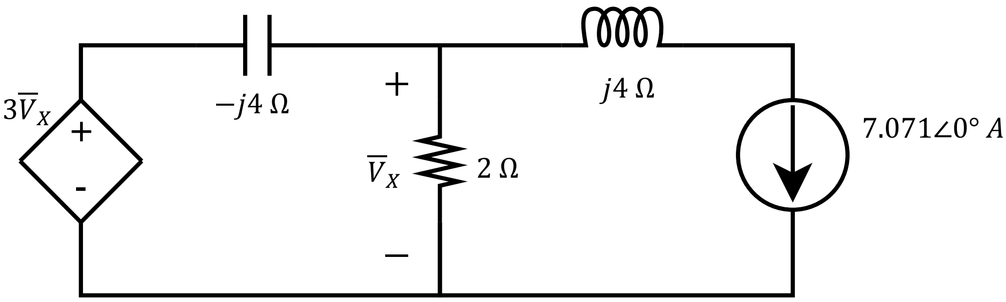
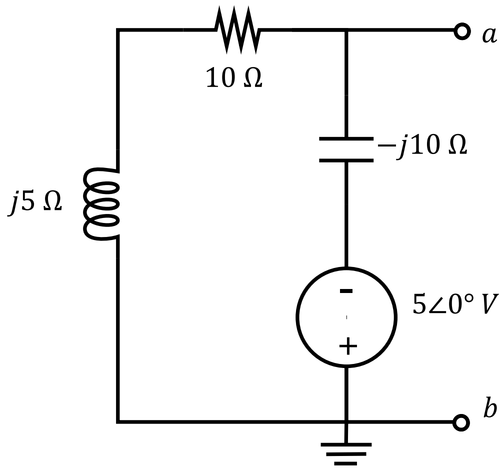
Vraag 3 [4] Gebruik lus-analise en bepaal onderstaande komplekse koeffisïente: $a\bar{I}_1 + b\bar{I}_2 = 4$ $j9\bar{I}_1 + c\bar{I}_2 = d$ (Skryf koeffisïente in **reghoekige formaat**.)	**Question 3 [4]** Use mesh analysis to determine the complex coefficients below: $a\bar{I}_1 + b\bar{I}_2 = 4$ $j9\bar{I}_1 + c\bar{I}_2 = d$ (Write coefficients in **rectangular format**.)



Vraag 4 [2] Bepaal $\overline{V}_X$. Skryf antwoord in **polêre formaat**.	**Question 4 [2]** Determine $\overline{V}_X$. Write answer in **polar format**.



Vraag 5 [4] Bepaal die Thevenin ekwivalent vir onderstaande kringbaan met betrekking tot terminale *a* – *b*. (Skryf $\overline{V}_{TH}$ en Z_{TH} in **reghoekige formaat**.)	**Question 5 [4]** Determine the Thevenin equivalent for the circuit below with respect to terminals *a* – *b*. (Write $\overline{V}_{TH}$ and Z_{TH} in **rectangular format**.)



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ANSWER SHEET

Surname:		*Student number:*								
Initials:										

Tot:	**/15**

1	$v_0(t) =$	
2	$a)\ Z_{eq} =$ (rectangular)	$b)\ Z_{eq} =$ (rectangular)
3	$a =$ (rectangular)	$b =$ (rectangular)
	$c =$ (rectangular)	$d =$ (rectangular)
4	$\overline{V}_X =$ (polar)	
5	$\overline{V}_{TH} =$ (rectangular)	$Z_{TH} =$ (rectangular)